

Self-Locking Tape Measure

Reverse Engineering & Part Definition

Shiv Sastry · 2023 · Modeled in Onshape · 8 components, 1 assembly, 10 sheets

OVERVIEW

A commercially available self-locking tape measure was fully disassembled and re-defined from scratch. Every component was dimensioned by hand with a dial caliper, assigned nominal dimensions and tolerances, detailed on a 2D drawing, modeled as a 3D part, and reassembled in CAD to verify that the locking mechanism clears through its full range of motion.

No manufacturer drawings, specifications, or CAD data were referenced. All geometry in this package originates from physical measurement of the product.

METHOD

- 01
- Teardown: disassembled the product into its eight discrete components and recorded the assembly order and interfaces.
- 02
- Measurement: dimensioned every feature with a dial caliper reading to 0.001 in.
- 03
- Tolerancing: assigned nominal dimensions and a general tolerance sized to measurement repeatability and to the clearances the mechanism requires to function.
- 04
- Detailing: produced a dimensioned 2D drawing for each component at a scale chosen for legibility (1:1 through 8:1).
- 05
- Modeling: built each component as a 3D solid from its own drawing.
- 06
- Assembly: mated all components, generated the exploded view and bill of materials, and checked the thumb lock through its range of motion for interference.

NOTES

1. All dimensions are in inches.

2. General tolerance: $\pm .005$ in unless otherwise noted. Selected to reflect the repeatability achievable measuring molded components with a hand caliper.

3. Materials in the bill of materials were identified by inspection of the physical product and are not verified against manufacturer specifications.

4. Mass values are calculated by the CAD system from the assigned materials.

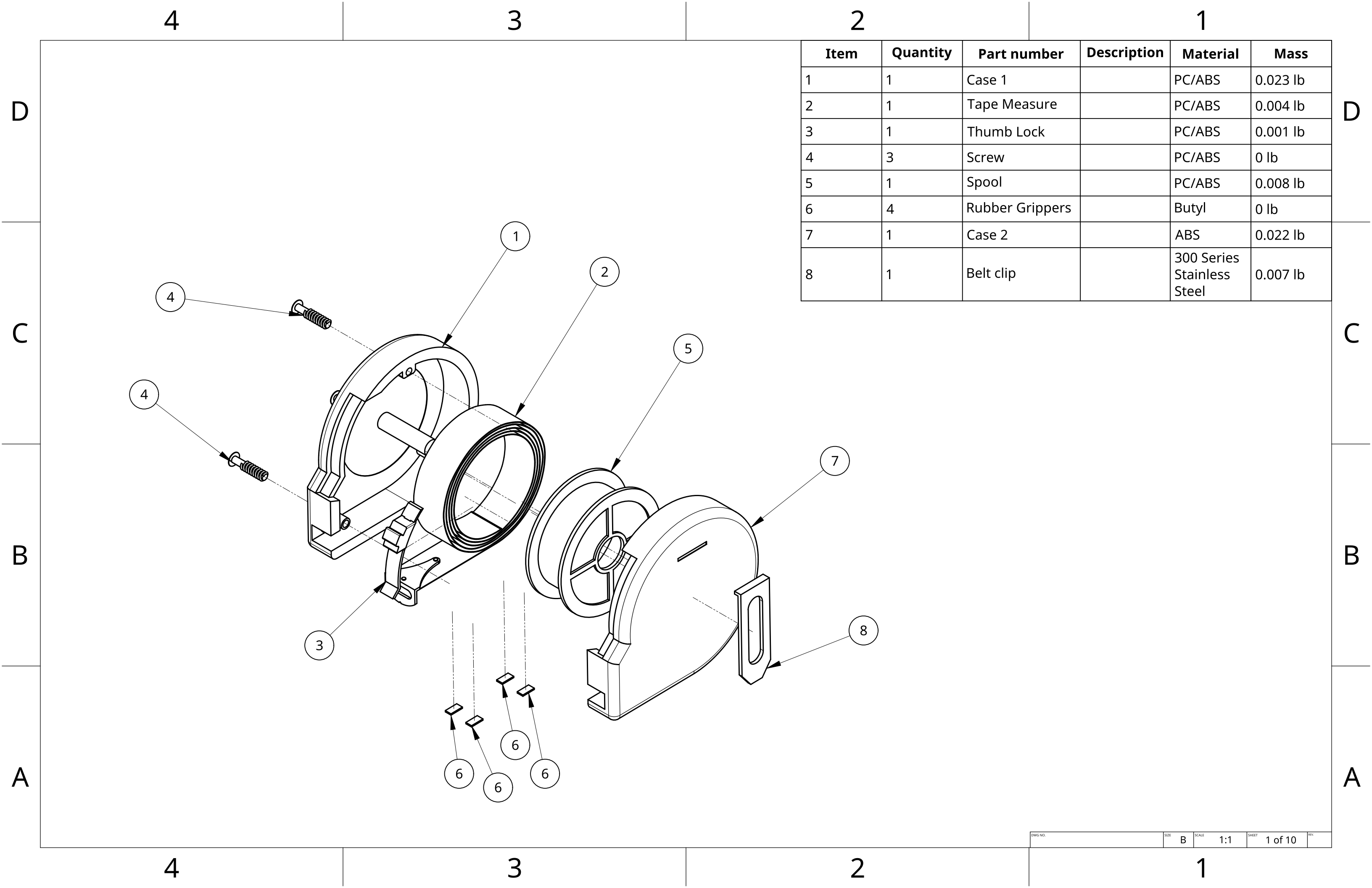
SHEET INDEX

SHEET	CONTENT	SIZE
1	Exploded assembly, ballooned, with bill of materials	B
2	General assembly, six views	B
3	Case 1, case half (spindle side)	B
4	Case 2, case half (belt-clip side)	B
5	Spool	B
6	Tape blade	B
7	Thumb lock	B
8	Screw (3 required)	B
9	Belt clip	A
10	Rubber gripper (4 required)	A

PRINCIPAL INTERFACES

INTERFACE	FIT REQUIRED
Spool bore on case spindle	Clearance, free rotation
Blade slot through case	Clearance, blade must feed and retract
Thumb lock in case guide	Clearance, sliding travel
Screws into case bosses	Interference, self-tapping
Grippers on case exterior	Bonded, no functional fit

TITLE	DRAWN BY	DATE	SIZE	UNITS	SHEET
Self-Locking Tape Measure	S. Sastry	2023	B	IN	Cover



Item	Quantity	Part number	Description	Material	Mass
1	1	Case 1		PC/ABS	0.023 lb
2	1	Tape Measure		PC/ABS	0.004 lb
3	1	Thumb Lock		PC/ABS	0.001 lb
4	3	Screw		PC/ABS	0 lb
5	1	Spool		PC/ABS	0.008 lb
6	4	Rubber Grippers		Butyl	0 lb
7	1	Case 2		ABS	0.022 lb
8	1	Belt clip		300 Series Stainless Steel	0.007 lb

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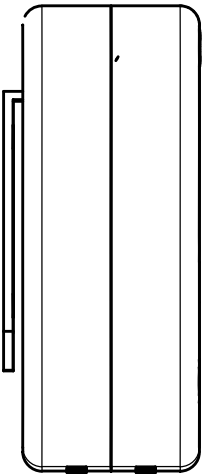
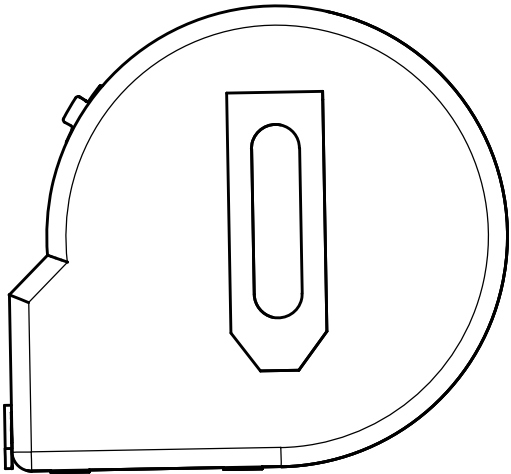
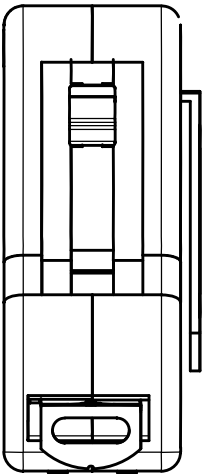
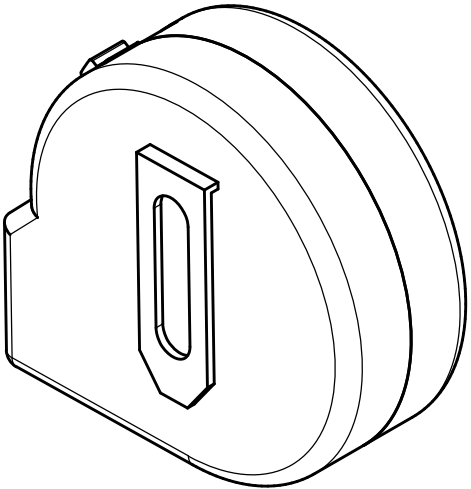
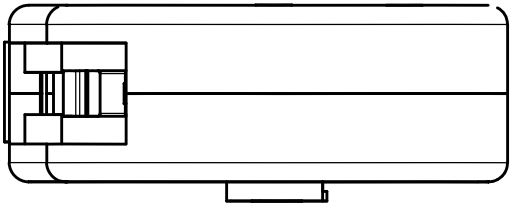
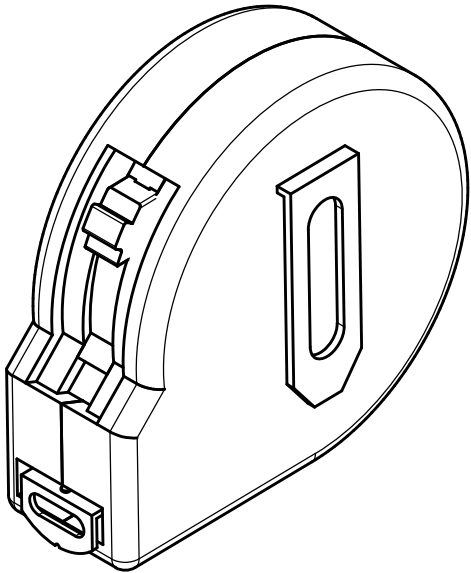
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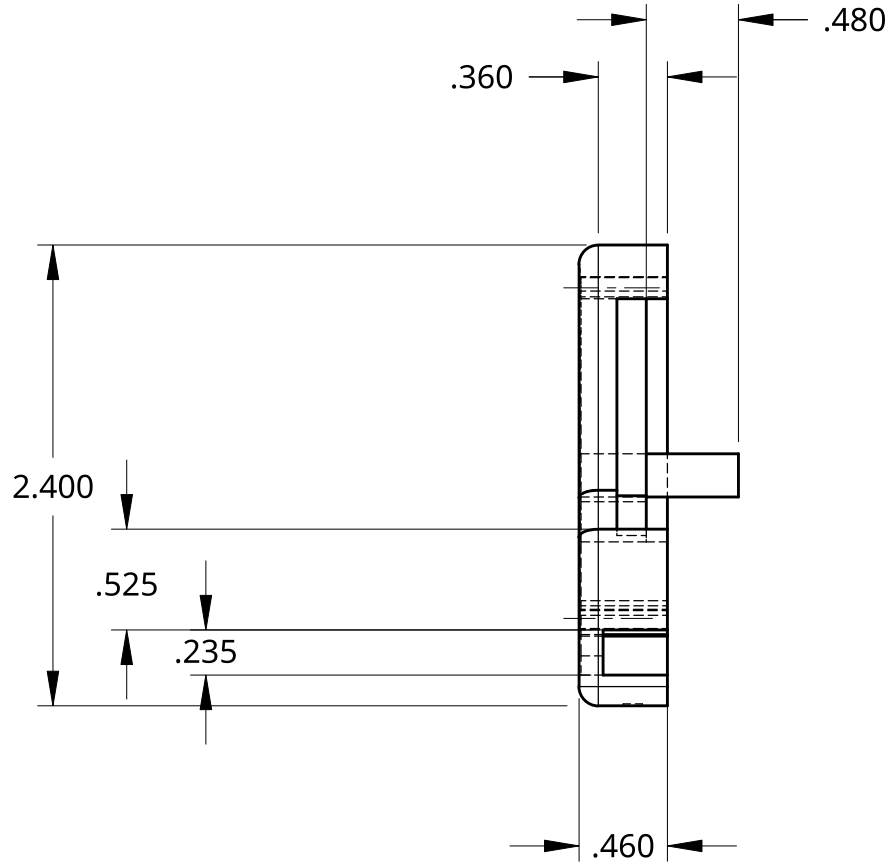
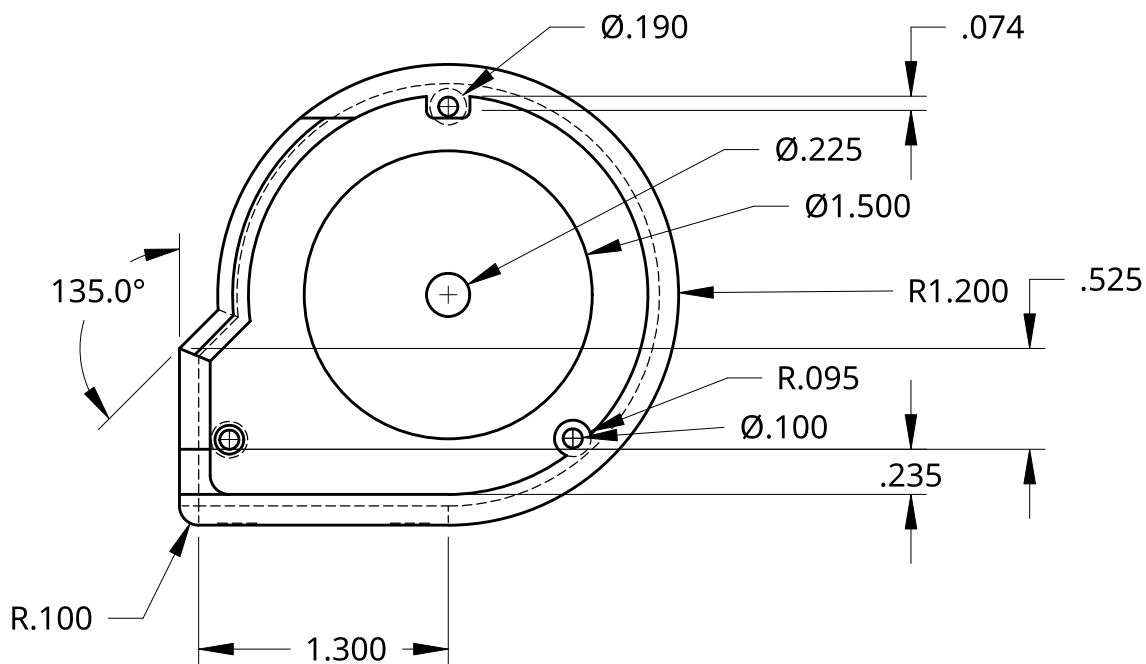
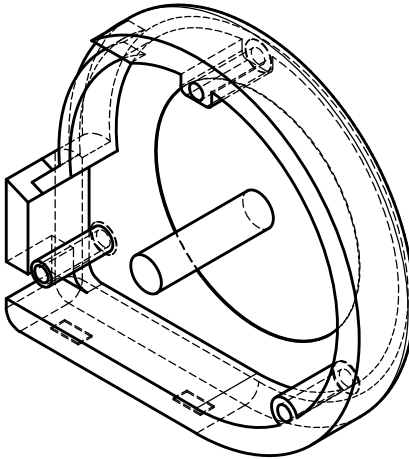
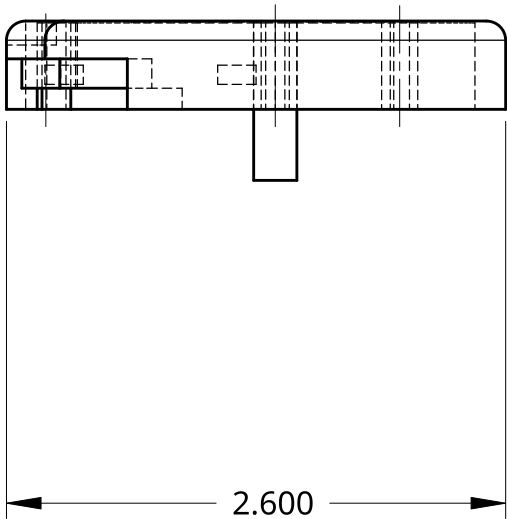
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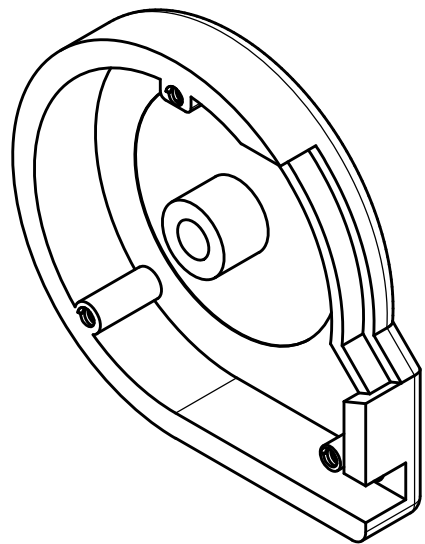
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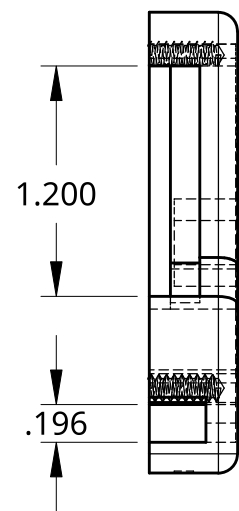
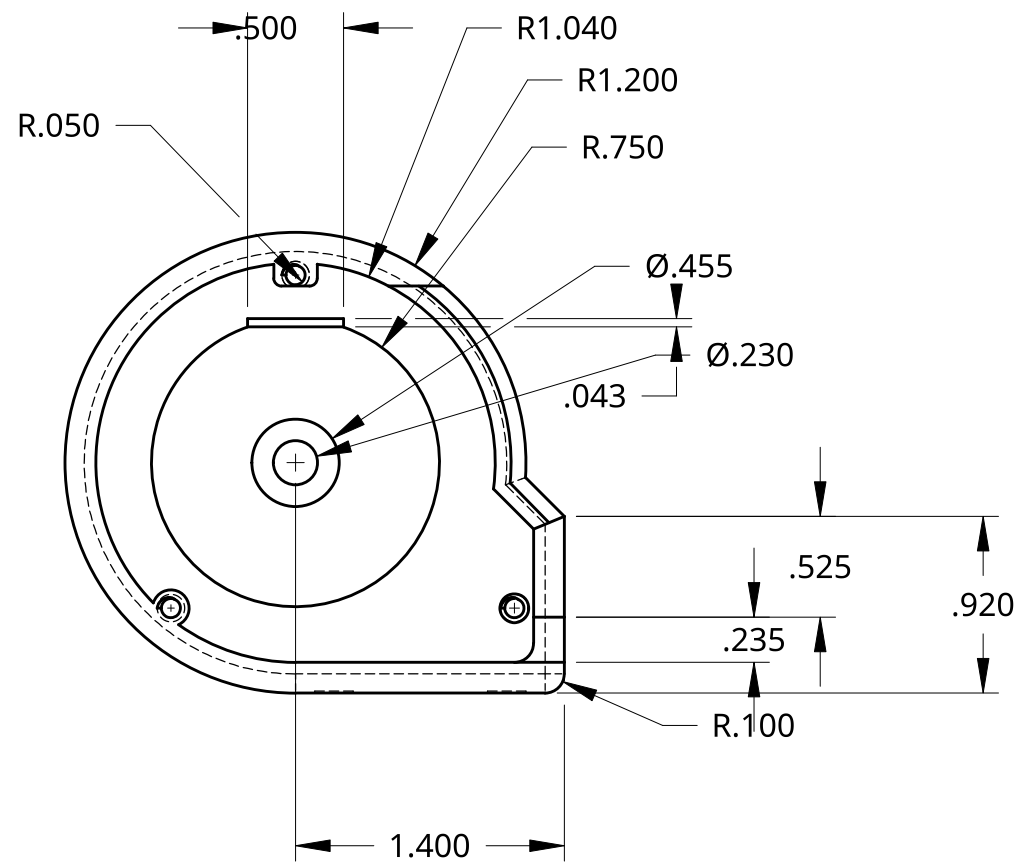
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DWG NO.	SIZE B	SCALE 1:1	SHEET 4 of 10	REV.
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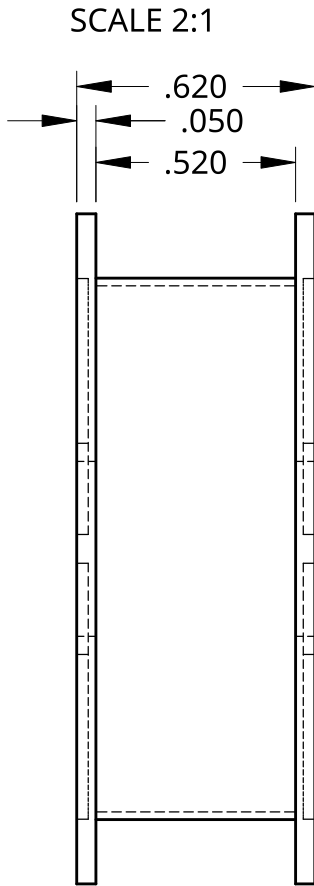
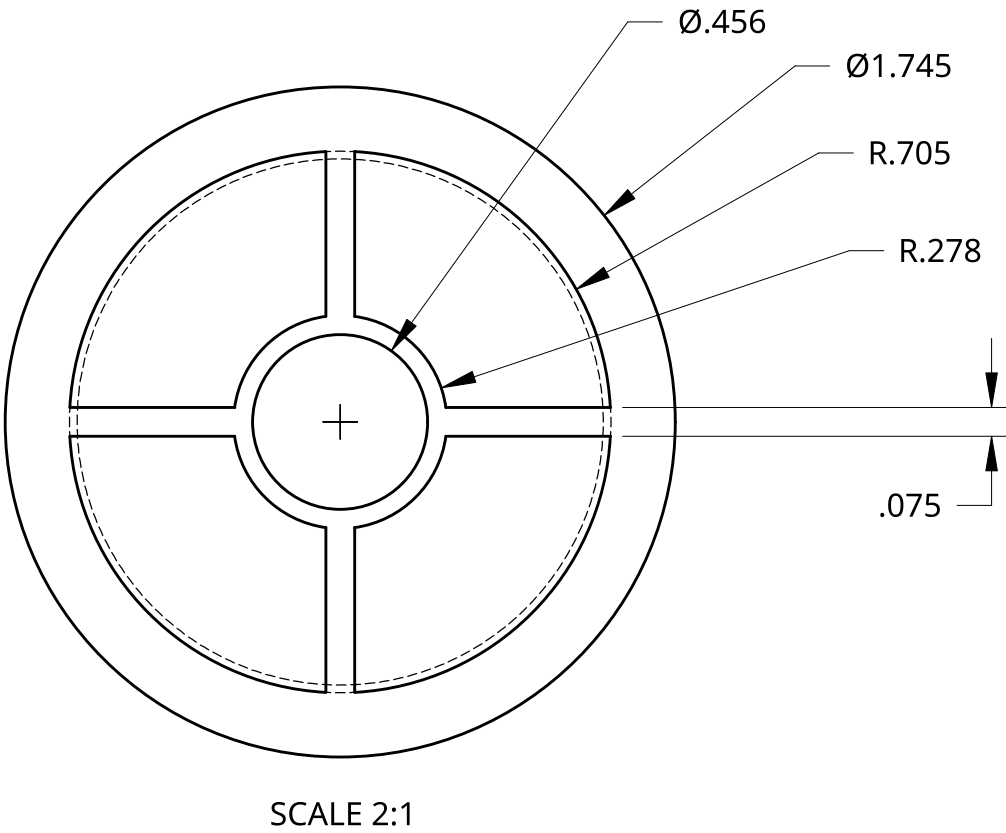
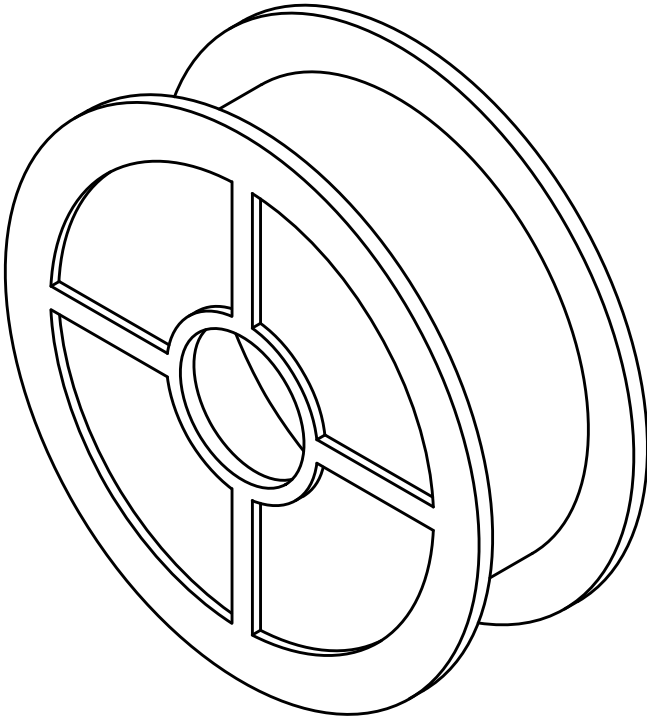
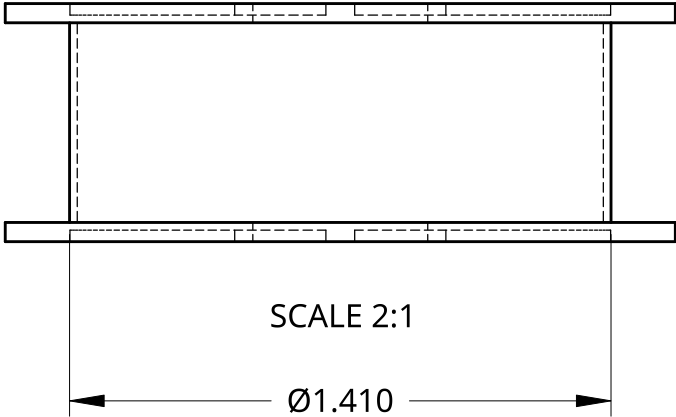
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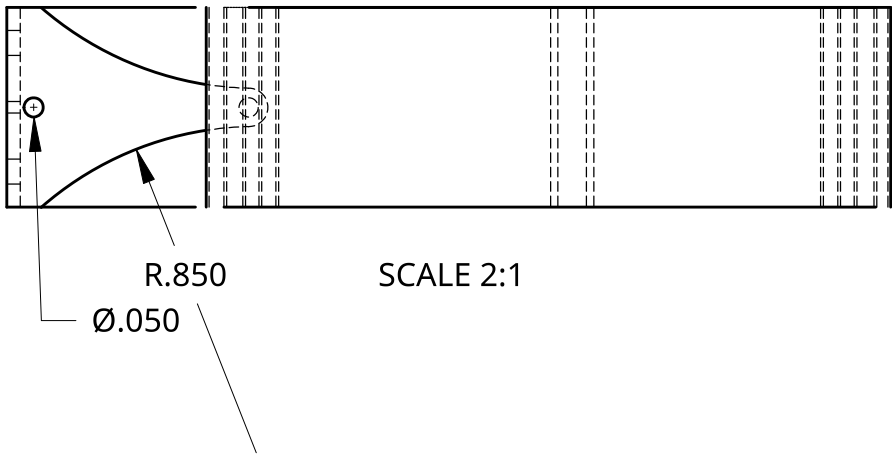
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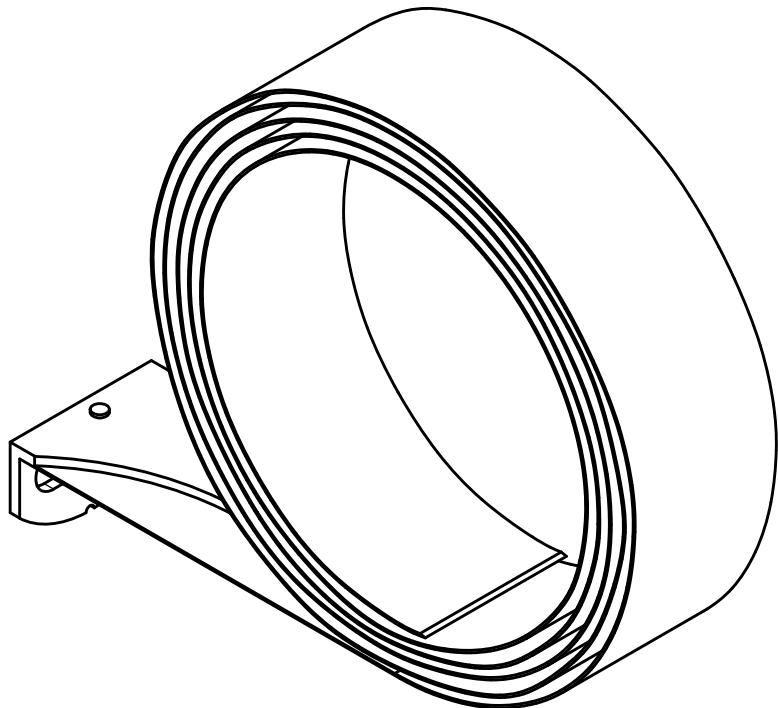
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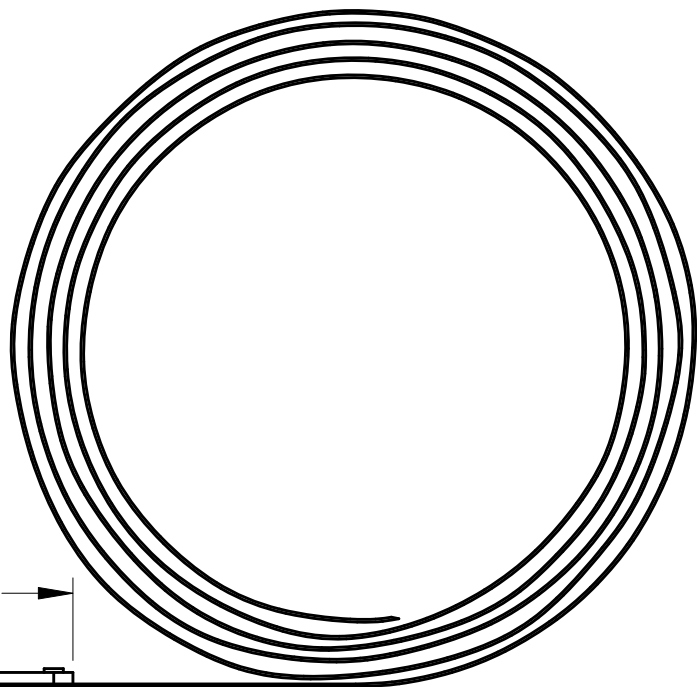
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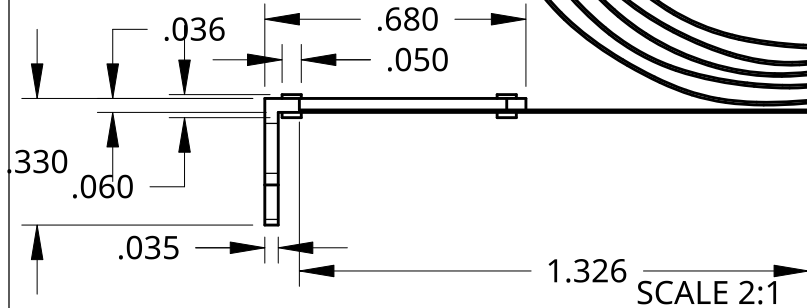
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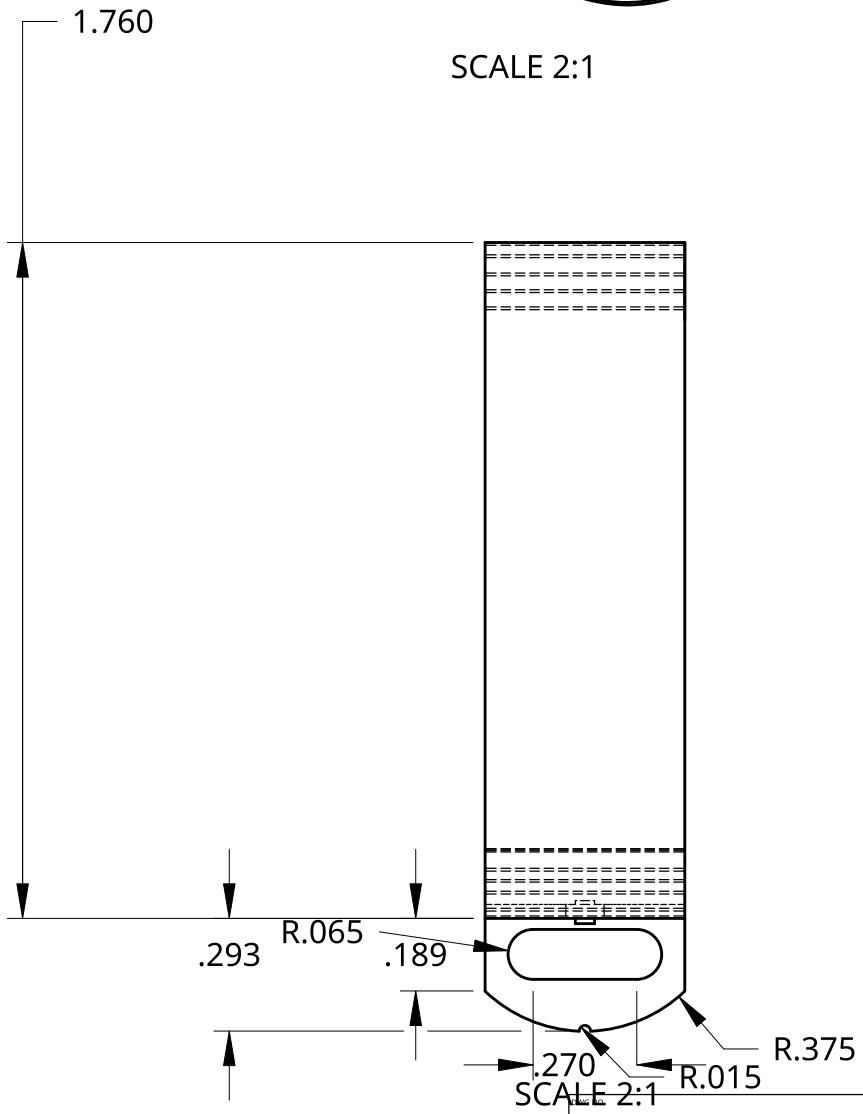


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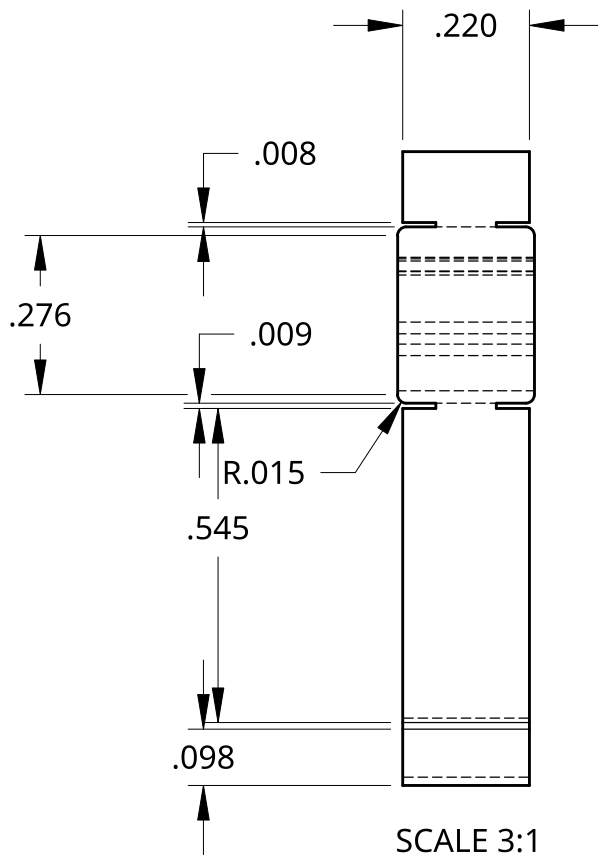
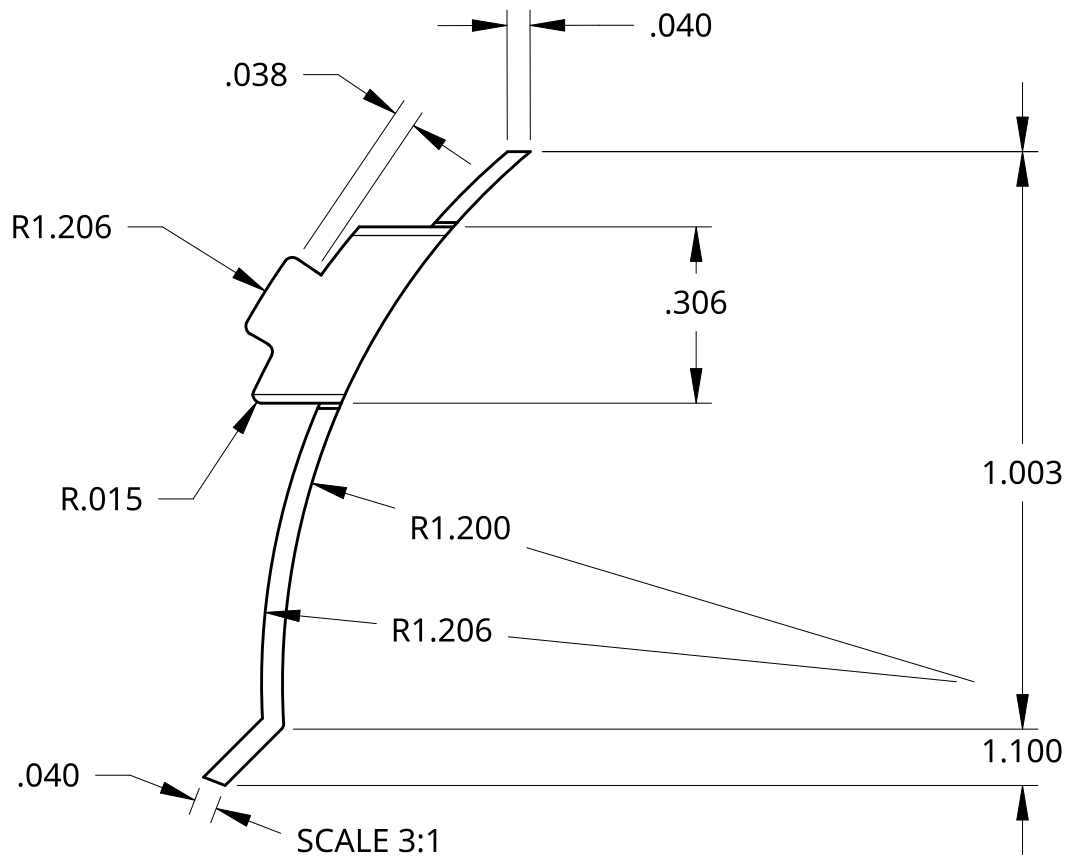
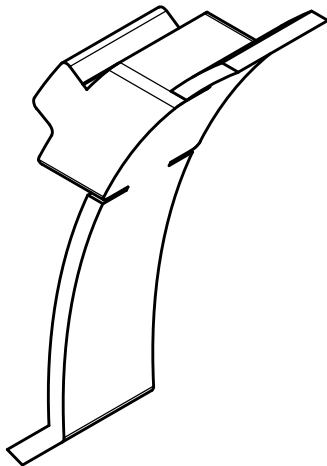
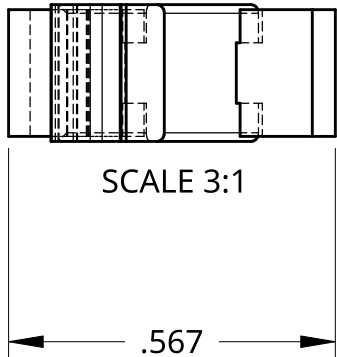
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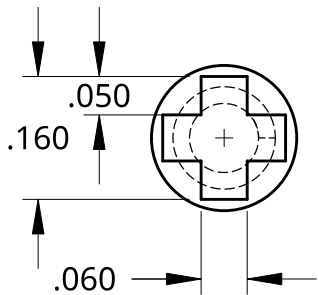
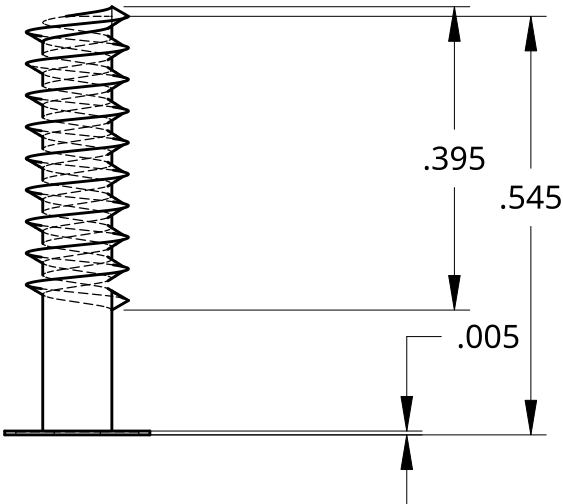
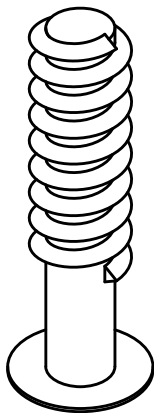
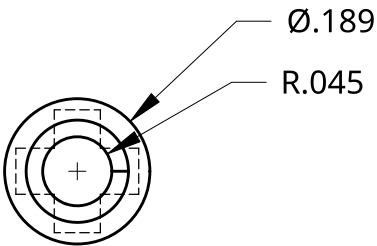
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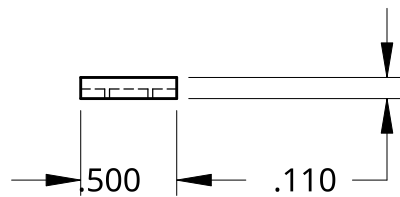
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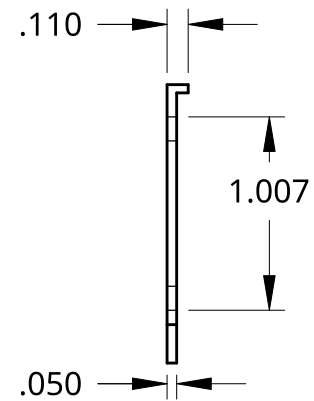
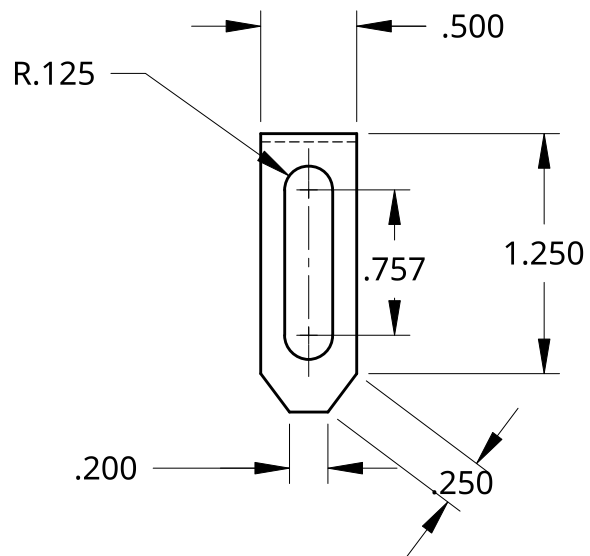
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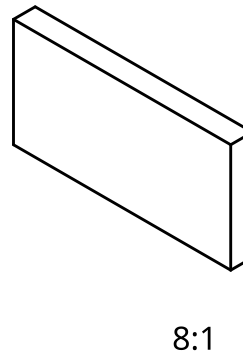
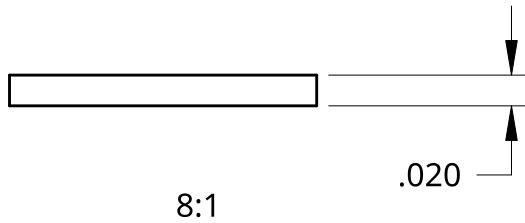
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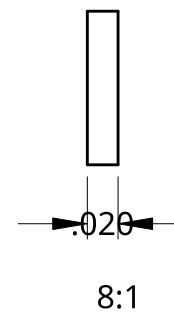
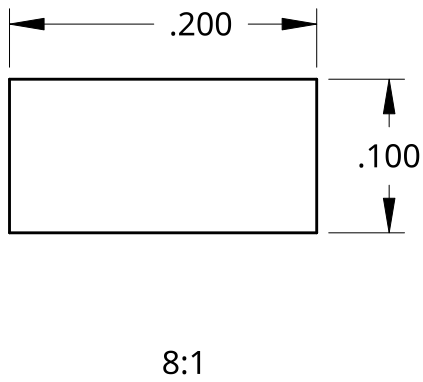
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